Shu LI

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Education & Research Experience

2026 **PhD in Computer Science**
ENS, PSL University & Inria Paris. Supervisors: Profs. Ana Bušić, Jean-Michel Fourneau
Thesis: Performance Paradoxes and Admission Control in Stochastic Matching Models.
Defended: Oct. 2026

2023 **MSc in Mathematics**
Shandong University of Technology. Supervisor: Prof. Jianfeng Wang
Thesis: On the Matrix-Tree Theorems of Graphs. Defended: May 2023

2019 **BSc in Mathematics**
Shandong University of Technology

Publications

2026 **Greedy failure and a stabilizing partial group discipline on a nested hypergraph**
S. Li, A. Bušić, and J.-M. Fourneau. ASMTA, Ghent, Belgium

2026 **State space truncation and admission control for matching systems**
A. Bušić, J.-M. Fourneau**, and S. Li**. NETGCOOP 2026, Palermo, Italy

2026 **Performance paradox in matching systems**
S. Li, A. Bušić, and J.-M. Fourneau. ROADEF, Tours, France

2026 **A graph discretization of vector Laplacian**
S. Li, L. Lu, and J. Wang. Discrete Appl. Math. 379, 446–460

2026 **Matrix-Tree Theorems for complex unit gain networks**
S. Li, J. Wang, M. Dehmer, M. Brunetti, and Y. Li. J. Comput. Appl. Math. 488, 117807

2026 **ORKM: an R package for online multi-view data clustering**
M. Yu**, S. Li**, and G. Guo. Neurocomputing, 663, 131973

2025 **Performance paradox in matching models with abandonment**
A. Bušić, J.-M. Fourneau**, and S. Li**. EAI ValueTools, Glasgow, Scotland

2025 **Performance paradoxes in matching systems are not that rare**
A. Bušić, J.-M. Fourneau, A. Lunven**, and S. Li**. NETGCOOP, Bilbao, Spain

2023 **Yet more elementary proof of Matrix-Tree Theorem for signed graphs**
S. Li and J. Wang. Algebra Colloq. 30(3), 493–502

2023 **Matrix-Tree Theorem of digraphs via signless Laplacians**
S. Li, L. Lu, J. Wang, and W. Wang. Linear Algebra Its Appl. 658, 32–48

2023 **On graphs with small ranks: old and new results**
S. Li, Z. Stanić, and J. Wang. Adv. Math. 52(3), 385–405

2017 **Equivalence relation and function**
S. Li and H. Li. Int. J. Trend Res. Dev. 4(6)

Preprints & Manuscripts

Submitted **On the spectral radius of trees with fixed eccentric sequence**
S. Li, M. Yu, and Y. Özer

2023 **A graph discretization of vector Laplace operator**
S. Li, L. Lu, and J. Wang, arXiv

Awards

2026 Best Paper Award, NETGCOOP 2026: *State space truncation and admission control for matching systems.*

2026 Excellence in Reviewing (2025–2026), Journal of Advances in Mathematics and Computer Science.

Presentations

2026 **Greedy failure and a stabilizing partial group discipline on a nested hypergraph**
The 29th International Conference on Analytical and Stochastic Modelling Techniques and Applications (ASMTA 2026), Ghent, Belgium

2026 **Steady-state performance paradoxes in stochastic matching networks with abandonment**
LINCS Annual Workshop with the Scientific Committee, Palaiseau, France

2026 **Performance paradox in matching systems**
27ème Congrès Annuel de la Société Française de Recherche Opérationnelle et d'aide à la Décision (ROADEF2026), Tours, France

2025 **Performance paradox in matching models with abandonment**
18th EAI International Conference on Performance Evaluation Methodologies and Tools (EAI VALUETOOLS2025), Glasgow, Scotland

2025 **Performance paradoxes in matching systems are not that rare**
The 12th International Conference of Networks, Games, Control and Optimization (NETGCOOP2025), Bilbao, Spain

2025 **Performance paradoxes in matching systems are not that rare**
LINCS Annual Workshop with the Scientific Committee, Palaiseau, France

2024 **A Brief Survey on the Matrix-Tree Theorems of Graphs**
Argo Seminar, Paris, France

2023 **On the Matrix-Tree Theorems of Graphs**
The Third SDUT Postgraduate Academic Forum, Zibo, China

2022 **On graphs with small ranks: old and new results**
The 10th Int'l Combinatorics and Graph Theory Conference (CMS-CGT2022), Harbin, China

2022 **A Survey on Matrix-Tree Theorem of Graphs**
The 9th Int'l Symposium on Graph Theory and Combinatorial Algorithms (ORSC-GTCA2022), Hangzhou, China

Academic Service

2026–Present Reviewer, Mathematical Reviews (MathSciNet)

2025–Present Reviewer, Zentralblatt MATH (zbMATH)

2025–Present Journal of Advances in Engineering and Technology

Peer Review

Jrnl. Discrete Applied Mathematics / IEEE Internet of Things Magazine / Asian Journal of Mathematics and Computer Research / Mathematics and Computer Science / Journal of Advances in Mathematics and Computer Science

2026 Confs. ICECET / ACDSA / INCOSST

2025 Confs. ICECET / GPS / ICIC / ACDSA / ICECCME

Professional Memberships

2023–Present International Linear Algebra Society (ILAS) / Société Mathématique de France (SMF) / Société informatique de France (SiF) / Société de Mathématiques Appliquées et Industrielles (SMAI) / Laboratory for Information, Networking and Communication Sciences (LINCS)

Research Software

Hypergraph Stochastic Matching *(Python)*
Simulation of matching policies with paired experiments, conservation checks, and reproducibility metadata.

Performance Paradoxes in Matching Networks *(Python, NetworkX)*
Numerical evaluation of stability conditions, bottleneck sets, and mean queue lengths in stochastic matching networks.

State-Space Truncation for Matching Systems *(Python)*
Exact rational computations for the paw-graph example in the NETGCOOP 2026 paper on truncation and admission control.

Matching Models with Abandonment *(Python)*
Randomized searches for performance paradoxes and numerical evaluation of stationary quantities with truncation-error bounds.

Spectral Radius and Eccentric Sequences *(Python, NetworkX)*
Enumeration of trees by eccentric sequence, spectral-radius comparisons, and visualization of extremal candidates.

Research Profile

I successfully defended my Ph.D. thesis in Computer Science in October 2026, following doctoral research at ENS and Inria Paris. My research focuses on stochastic matching models within applied probability and stochastic networks, particularly performance paradoxes and admission control.

Before my doctoral studies, I earned an M.Sc. in Mathematics with a focus on graph theory, studying Matrix-Tree Theorems, low-rank graph families, and the vector Laplacian on graphs. I also hold a B.Sc. in Mathematics and Applied Mathematics.

My mathematical training provides a strong foundation in linear algebra, number theory, and modern algebra. Beyond my thesis, I am interested in connecting theoretical insights with practical problems in computer science and mathematics.